Figure 1A: Macroevolutionary Locus Divergence Across Key Clades & Extrema (n=33 Species)

A. Whole-Gene / Isochore GC Z-Scores (Chromosomal Domain Shift)

B. Coding Sequence (CDS) GC Z-Scores (Globin Coding Divergence)

Human (H. sapiens)
Sperm Whale (P. catodon)
Alpaca (V. pacos)
Elephant Seal (M. angustirostris)
Vampire Bat (D. rotundus)
Mammalia
Naked Mole-Rat (H. glaber)
Mouse (M. musculus)
Elephant (L. africana)
Opossum (M. domestica)
Echidna (T. aculeatus)
Monotremata
Platypus (O. anatinus)
Golden Eagle (A. chrysaetos)
Chicken (G. gallus)
Aves
King Penguin (A. patagonicus)
Alligator (A. mississippiensis)
Archosauria
Adder (V. berus)
Wall Lizard (P. muralis)
Anole (A. carolinensis)
Reptilia
Sea Turtle (C. mydas)
Painted Turtle (C. picta)
Clawed Frog (X. tropicalis)
Axolotl (A. mexicanum)
Amphibia
Gabon Caecilian (G. seraphini)
Coelacanth (L. chalumnae)
Sarcopterygii
Fugu (T. rubripes)
Zebrafish (D. rerio)
Teleostei
Spotted Gar (L. oculatus)
Bichir (P. senegalus)
Holostei
Grey Reef Shark (C. amblyrhynchos)
Catshark (S. canicula)
Chondrichthyes
Greenland Shark (S. microcephalus)
Sea Lamprey (P. marinus)
Agnatha
Inshore Hagfish (E. burgeri)

α-Globin Locus (Mean Z)
β-Globin Locus (Mean Z)
Divergence: α > β
Inversion: β > α
Flanking Syntenic Non-Globins
Genome Typical Range (±1σ)

−5 −4 −3 −2 −1 0 1 2 3 4 5 6

Standardized Z-Score: Whole-Gene GC (σ from Genome Baseline)

α-Globin CDS (Mean Z)
β-Globin CDS (Mean Z)
Divergence: α > β
Inversion: β > α

−2 −1 0 1 2 3

Standardized Z-Score: CDS GC (σ from Genome Baseline)